DSA C++ Exam MCQs (All Lectures)

Lecture 1: Introduction to DSA

1. Which of the following is a characteristic of an algorithm?

- A) It must be written in C++
- B) It must have ambiguous steps
- C) It must be finite and well-defined ✓
- D) It must execute in O(1) time

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2. Big O notation describes:

- A) Best case scenario
- B) Average case scenario
- C) Worst case scenario ✓
- D) Exact runtime

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3. Which is NOT a basic operation in algorithm analysis?

- A) Arithmetic operation
- B) Comparison
- C) Assignment
- D) User input prompt ✓

Lecture 2: Arrays

4. What is the time complexity of accessing an element by index in an array?

- A) O(n)
- B) O(1) ✓
- C) O(log n)
- D) O(n^2)

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5. Which operation is most expensive in a static array?

- A) Accessing element at index 5
- B) Inserting element at beginning ✓
- C) Deleting last element
- D) Updating element at index 10

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6. The main disadvantage of arrays is:

- A) Fast access time
- B) Fixed size ✓
- C) Contiguous memory
- D) Simple implementation

Lecture 3: Linked Lists

7. In a singly linked list, each node contains:

- A) Data only
- B) Data and pointer to next node ✓
- C) Data and two pointers
- D) Pointer only

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8. Insertion at beginning of singly linked list has time complexity:

- A) O(n)
- B) O(1) ✓
- C) O(log n)
- D) $O(n^2)$

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9. Advantage of linked list over array:

- A) Faster random access
- B) Dynamic size ✓
- C) Less memory usage
- D) Cache friendly

Lecture 4: Circular Linked Lists

10. In circular linked list, last node points to:

- A) NULL
- B) First node ✓
- C) Previous node
- D) Head node only

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11. Circular linked list is useful for:

- A) Binary search
- B) Round-robin scheduling ✓
- C) Quick sort
- D) Recursive algorithms

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12. In circular linked list with tail pointer, deletion of first node takes:

- A) O(1) time ✓
- B) O(n) time
- C) O(log n) time
- D) $O(n^2)$ time

Lecture 5: Stacks

13. Stack follows which principle?

- A) FIFO (First In First Out)
- B) LIFO (Last In First Out) ✓
- C) Random access
- D) Priority based

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14. Which application typically uses stack?

- A) Printer queue
- B) Function call management ✓
- C) CPU scheduling
- D) File system

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15. Postfix expression "5 3 + 2 *" evaluates to:

- A) 13
- B) 16 ✓
- C) 10
- D) 8

Lecture 5B: More on Stacks

16. Stack overflow occurs when:

- A) Stack is empty
- B) Trying to pop from empty stack
- C) Stack exceeds its capacity ✓
- D) Memory leak happens

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17. Which is NOT a stack operation?

- A) Push
- B) Pop
- C) Peek
- D) Enqueue ✓

Lecture 6: Queues

18. Queue follows which principle?

- A) LIFO (Last In First Out)
- B) FIFO (First In First Out) ✓
- C) Both LIFO and FIFO

- D) None of these

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19. In circular queue, front == rear indicates:

- A) Queue is full
- B) Queue is empty ✓
- C) Queue has one element
- D) Invalid state

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20. Dequeue operation removes from:

- A) Front ✓
- B) Rear
- C) Middle
- D) Any position

Lecture 6: Recursion

21. Recursion must have:

- A) Loop structure
- B) Base case ✓
- C) Multiple returns

- D) Global variables

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22. Which problem is best solved recursively?

- A) Linear search
- B) Fibonacci series ✓
- C) Array traversal
- D) Simple calculations

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23. Recursion uses which data structure implicitly?

- A) Queue
- B) Array
- C) Stack ✓
- D) Linked list

Lecture 7: Trees

24. Tree with n nodes has how many edges?

- A) n
- B) n-1 ✓
- C) n+1

- D) n/2

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25. Which is NOT a tree traversal?

- A) Inorder
- B) Preorder
- C) Postorder
- D) Sideorder ✓

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26. Full binary tree has:

- A) 0 or 2 children per node ✓
- B) Only left children
- C) Only right children
- D) Exactly 2 children always

Lecture 8: Balanced Trees

27. AVL tree is:

- A) Unbalanced BST
- B) Height-balanced BST ✓
- C) Weight-balanced tree

- D) Complete binary tree

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28. Balance factor in AVL tree is difference between:

- A) Left and right subtree heights ✓
- B) Number of left and right nodes
- C) Depth and height
- D) Parent and child values

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29. Which rotation for left-left imbalance?

- A) Left rotation
- B) Right rotation ✓
- C) Left-right rotation
- D) Right-left rotation

Lecture 9: Heap

30. Max-heap property:

- A) Parent ≤ children
- B) Parent ≥ children ✓
- C) Left child > right child

- D) Sorted array

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31. Heap is typically implemented using:

- A) Linked list
- B) Array ✓
- C) Stack
- D) Queue

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32. Time to get max from max-heap:

- A) O(n)
- B) O(1) ✓
- C) O(log n)
- D) O(n log n)

Lecture 10: Graphs

33. BFS uses which data structure?

- A) Stack
- B) Queue ✓
- C) Heap

- D) Array

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34. DFS uses which data structure?

- A) Stack ✓
- B) Queue
- C) Heap
- D) Tree

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35. Complete graph with n vertices has how many edges?

- A) n
- B) n(n-1)/2 ✓
- C) n-1
- D) n^2

Mixed/Advanced

36. Which sorting uses divide and conquer?

- A) Bubble sort
- B) Merge sort ✓
- C) Selection sort

- D) Insertion sort

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37. Hash table collision can be resolved by:

- A) Chaining ✓
- B) Sorting
- C) Recursion
- D) Traversal

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38. Dijkstra's algorithm finds:

- A) Minimum spanning tree
- B) Shortest path ✓
- C) Maximum flow
- D) Hamiltonian path

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39. In adjacency matrix, time to check edge between u and v:

- A) O(1) ✓
- B) O(n)
- C) O(log n)
- D) $O(n^2)$

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40. Time complexity of BFS on graph with V vertices, E edges:

- A) O(V)
- B) O(E)
- C) O(V+E) ✓
- D) O(V^2)

ANSWER KEY: All correct answers are marked with ✓ in the options above.